DuQuantum Pasqal Challenge

Quantum Optimization with Pasqal neutral atoms

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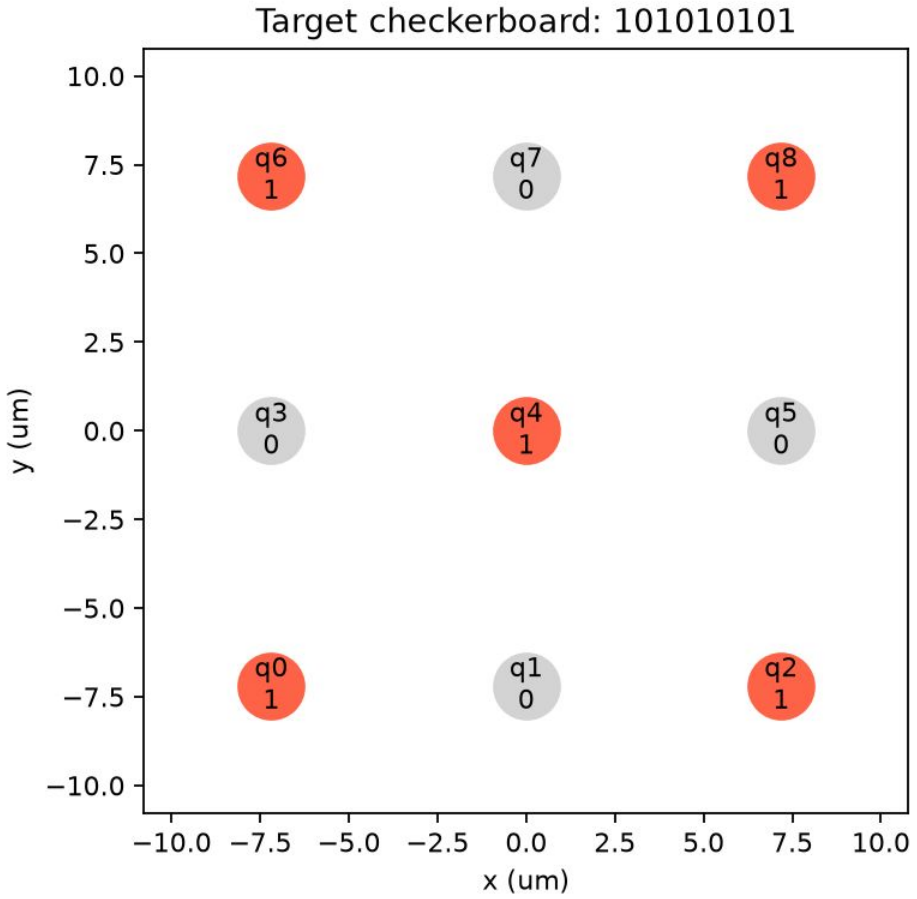
Pasqal and neutral-atom optimization

Analog quantum processors from neutral atoms held in optical traps.

Lasers drive atoms between a ground state and an excited Rydberg state.

Rydberg interactions allow the system to enter a ground state mimicking the ideal

Pulser specifies the register and pulses, then samples binary outcomes.



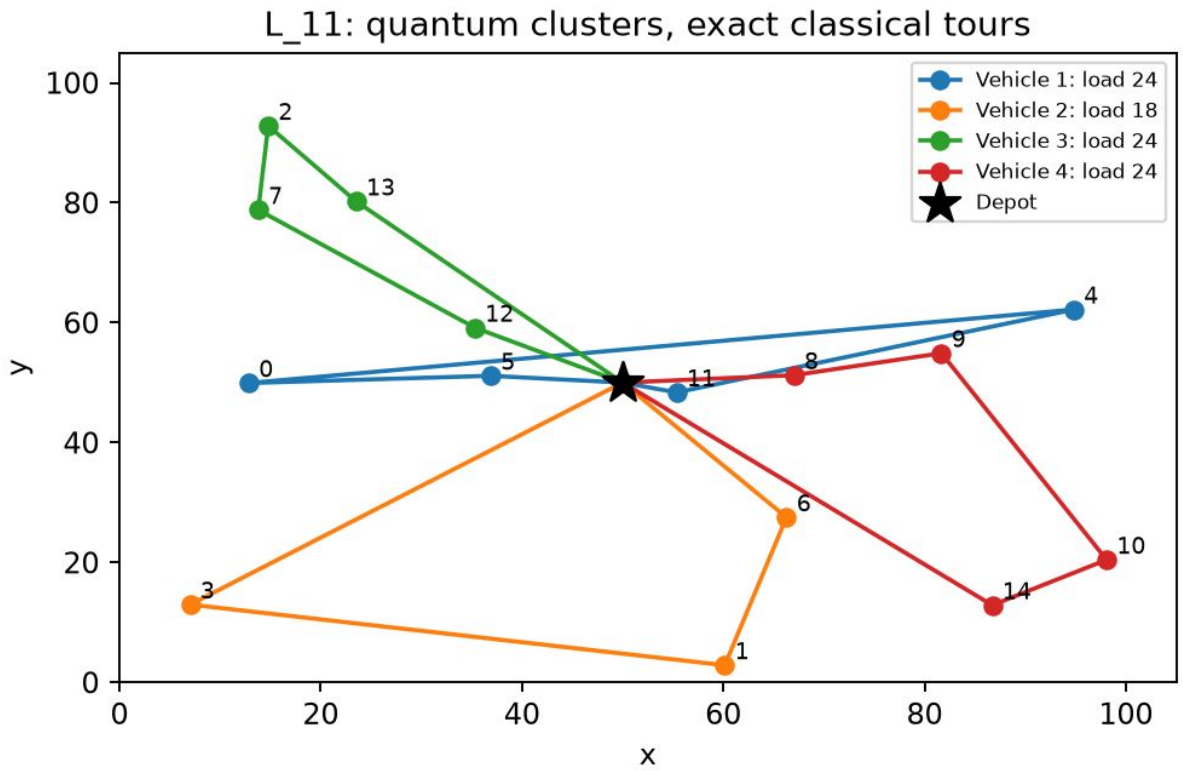
Saved nine-atom tutorial register; nearest-neighbor spacing 7.19 μm.

The capacitated vehicle routing problem

Tier	Customers	Vehicles	Capacity
S	4	2	7
M	10	3	20
L	15	4	30

Minimize total depot-to-depot distance.

Serve every customer once.
Respect each vehicle’s capacity.
Allow unused vehicles.



Saved L_{11} example: quantum clustering with exact classical tours

QUBO formulation and capacity slack

Binary decisions become an energy

$$E(\mathbf{b}) = c_0 + \sum_i h_i b_i + \sum_{i<j} J_{ij} b_i b_j, \qquad b_i \in \{0, 1\}$$

Linear terms score selections. Pair terms score joint choices.

Slack represents unused vehicle capacity

$$3x_1 + 2x_2 + s = 7, \qquad s = u_0 + 2u_1 + 4u_2$$

Load 5 + unused capacity 2 = capacity 7

The exact slack model is a reference. Our executed sampler checks capacity classically.

The implemented hybrid solver

Propose the next group

$$E_{\mathrm{cl}}/\kappa = \lambda m^2 - \mu m + \sum_{i<j} w_{ij} b_i b_j$$

$$m = \sum_i b_i, \qquad \mu = 2\lambda \frac{|R|}{L}$$

At most 8 candidate customers per call.
Group size is centered near |R|/L.
Classical checks reject overloaded proposals.

λ = 0.15; κ > 0 sets the energy scale.

Order each group's stops

$$E_{\mathrm{edge}} = \sum_e \tilde{c}_e e_e + A \sum_v \left(D_v(\mathbf{e}) - 2\right)^2$$

4 customers: 10 edge bits
or 16 position bits.

A strict decoder checks every sampled tour.

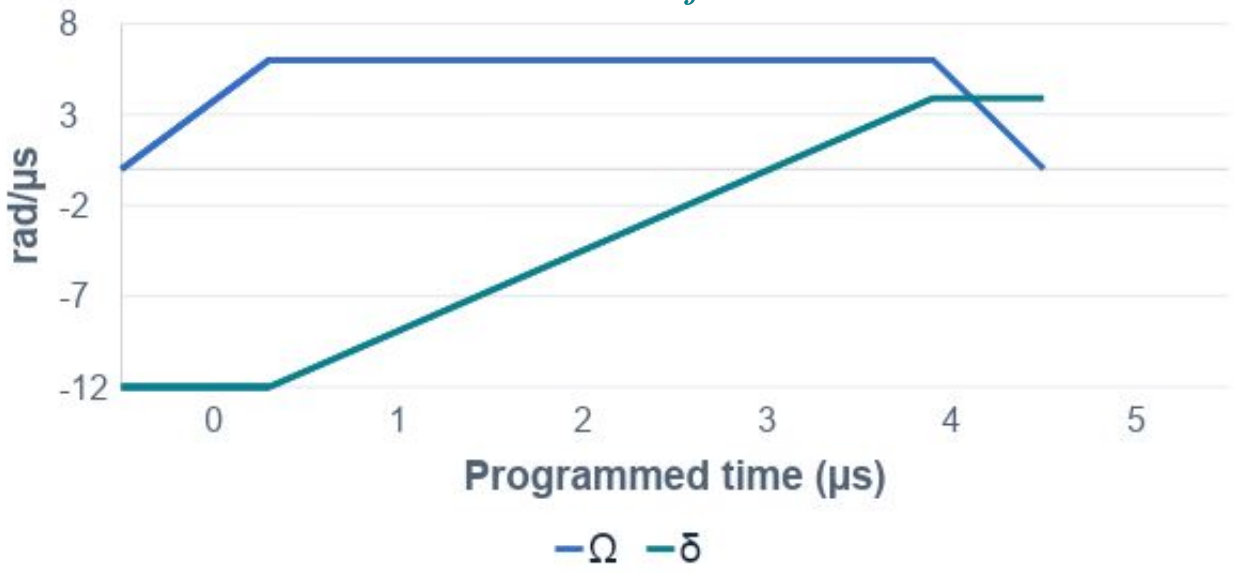
Degree = selected edges at a location.

Official benchmark: groups of ≤3 or >4 customers use classical routing. Atlas Orbit samples all three tours.

Mapping QUBO coefficients to atoms

$$\frac{H(t)}{\hbar} = \frac{\Omega(t)}{2}\sum_i X_i - \sum_i \delta_i(t) n_i + \sum_{i<j} \frac{\Gamma}{r_{ij}^6} n_i n_j$$

$$h_i^{\text{real}} = -\delta_i(T), \qquad J_{ij}^{\text{real}} = \frac{\Gamma}{r_{ij}^6}$$



Baseline pulse at 15 remaining customers / 4 vehicles

Detuning sets selection costs.

Atom separation sets pair costs.

The drive prepares the state.

5.2 µs pulse duration
Local CPU emulation; DMM adds local detuning

CPLEX leads on the 12-instance benchmark

Metric	CPLEX MILP	QUBO-assisted
Feasible full routes	36/36	33/36
Mean route gap*	0%	20.59%
Median total time	0.042 s	1.401 s
Proven optimal runs	36/36	0/36

12 instances × 3 repeats

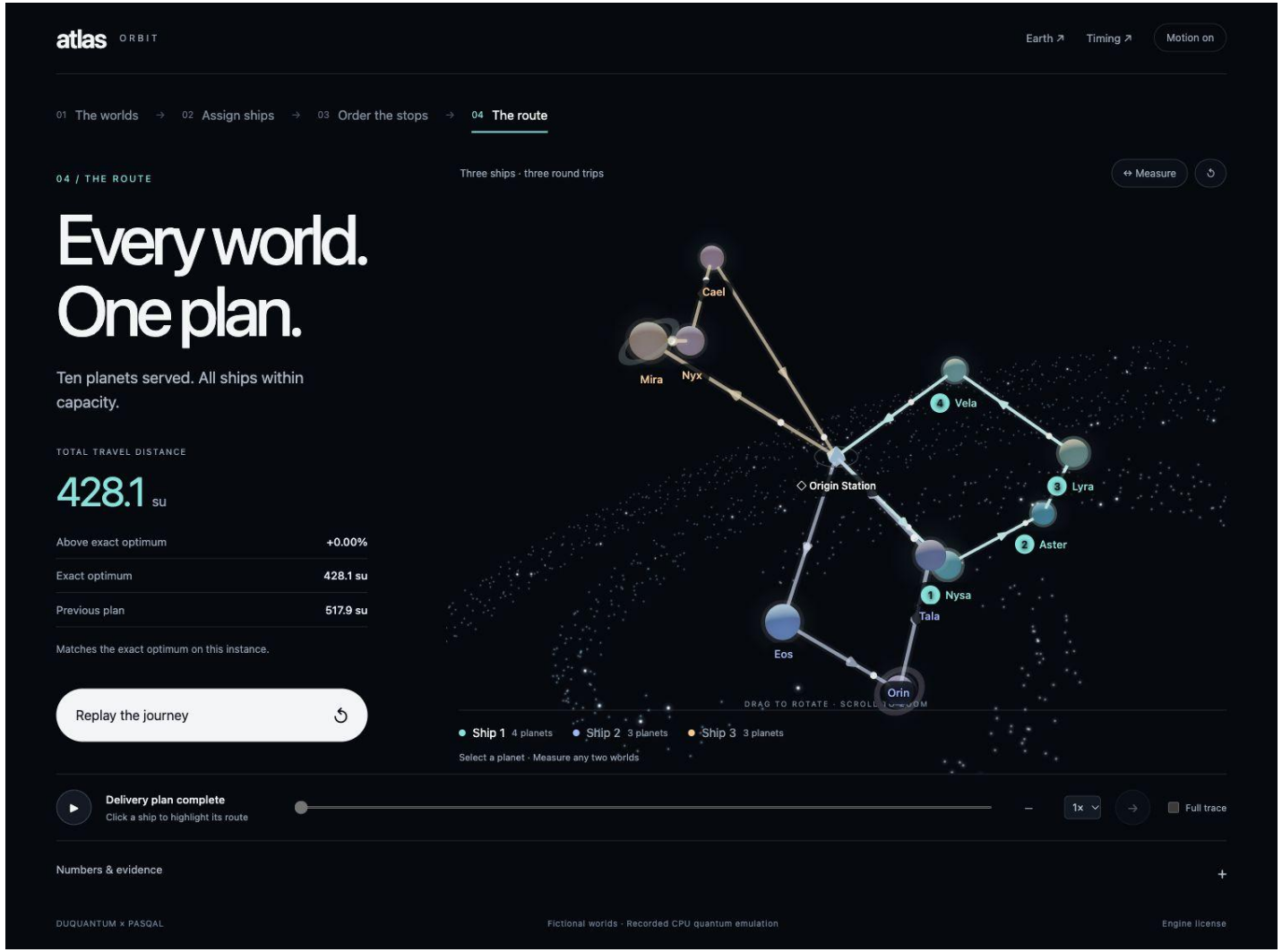
S/M/L, uniform and heterogeneous demands
Same routing objective and feasibility checks

797 / 797

target QUBO minima proven by CPLEX and checked by exhaustive enumeration

*Mean across 11 instances with successful emulator runs. CPU timings, no QPU. All 3 failures: M_heterogeneous_11.

Atlas Orbit matches the optimum on one instance



428.13 su

Matches the exact CVRP optimum

17.34% less travel

Compared with the prior 517.93 su demo run

All 3 tours sampled

10, 6 and 6 atoms
16,000 shots per ship

Fixed 3D demo. Larger budget: 512 clustering attempts, 4 window orders. CPU emulation; no speedup claim.

Where solution quality is lost

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fitted native models retain a target ground-state bitstring

Finite-time pulses can still sample useful excited states.

Clustering surrogate

Compact groups can still produce long depot tours.

Sequential capacity checks

A feasible early group can leave an infeasible remainder.

Physical embedding

Geometry distorts pair costs, even with local detuning.

What we built and what comes next

A reproducible hybrid routing experiment

QUBO derivations, measured bitstrings, strict route validation, and independent classical comparisons

Next: improve the model and embedding, then retest across more seeds

The updated Atlas Orbit demo reaches its exact optimum.

One larger-budget demo succeeds. CPLEX remains stronger on the 12-instance benchmark.

Exact assignment model and variable budget

$$E_{\text{slack}} = F(x) + A\sum_i \left(\sum_k x_{ik} - 1\right)^2 + B\sum_k \left(\sum_i d_i x_{ik} + s_k - Q\right)^2$$

$$s_k = \sum_{\ell=0}^{B_Q-1} 2^\ell u_{k\ell}, \qquad B_Q = \lceil \log_2(Q+1) \rceil$$

Encoding	S	M	L
Assignment x	8	30	60
Assignment + slack	14	45	80
Implemented window peak	4	8	8

A = B = U + 1 makes exact global minima feasible, given a feasible score U and nonnegative F.

Two routing encodings for four customers

Edge encoding: 10 binary variables

$$E_{\text{edge}} = \sum_{i<j} \tilde{c}_{ij} e_{ij} + A \sum_v \left(\sum_{e \ni v} e_e - 2 \right)^2$$

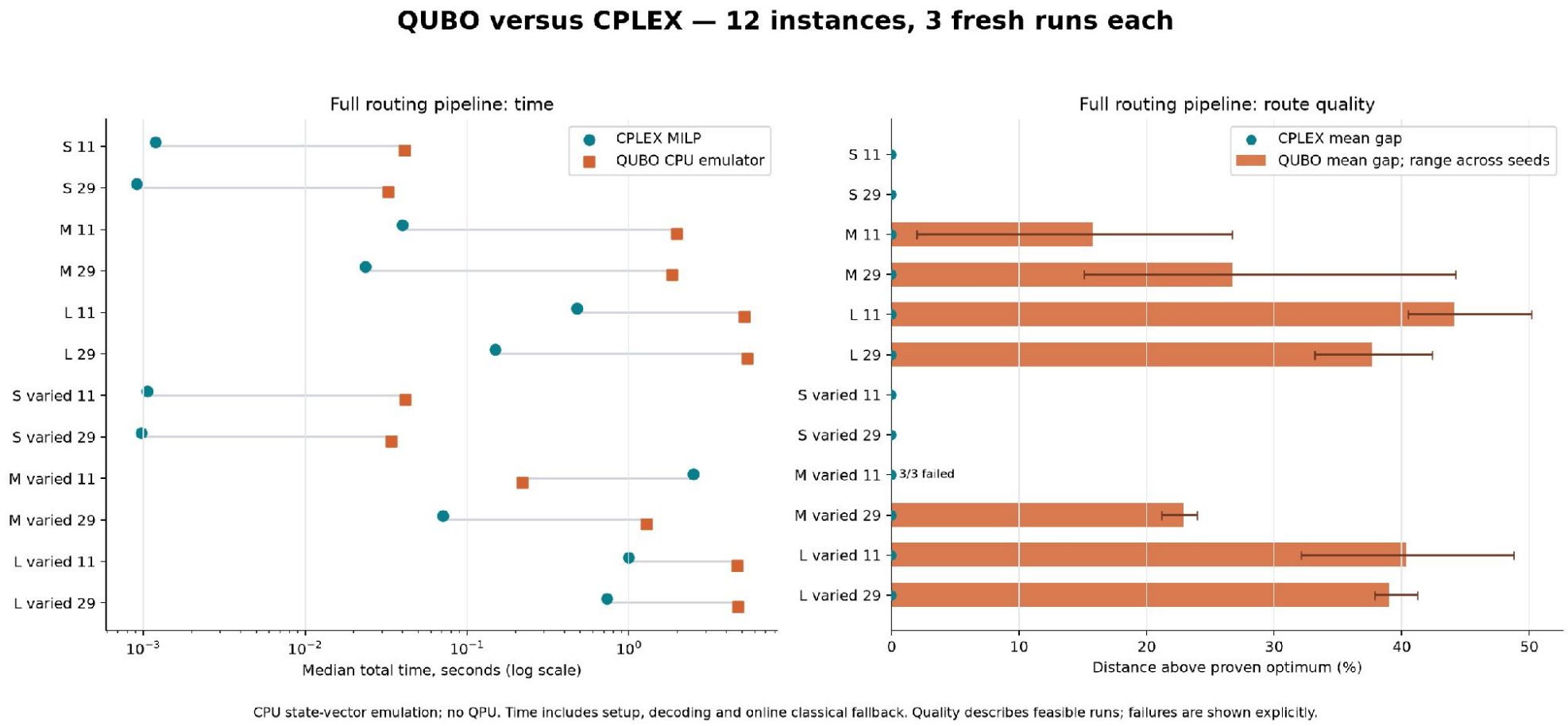
$$h_e = \tilde{c}_e - 6A, \qquad J_{ef} = 2A \quad \text{for incident edges}$$

Position encoding: 16 binary variables

$$E_{\text{pos}} = C_{\text{tour}}(x) + A \sum_i \left(\sum_t x_{it} - 1 \right)^2 + A \sum_t \left(\sum_i x_{it} - 1 \right)^2$$

Degree two implies one tour only for ≤4 customers here. Position bits encode a permutation. A = 10.

Classical comparison by instance



Local baseline runs, three repeats per instance. Failed solutions retain undefined route gaps.

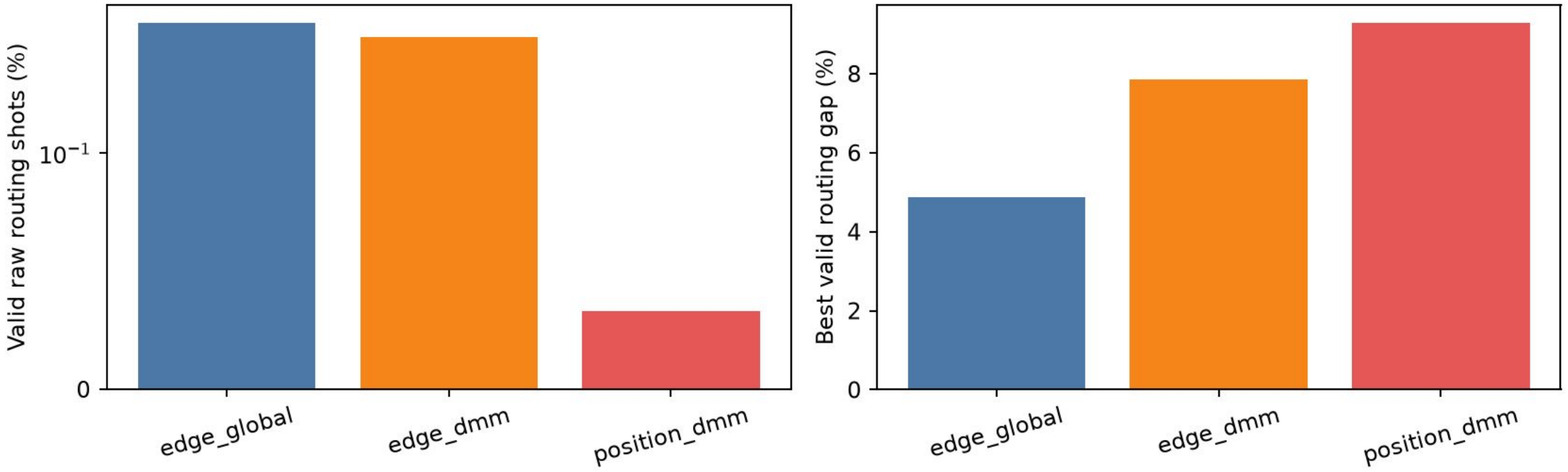
Identical-QUBO diagnostics

Stage / encoding	Banks	Sampled target optimum	Native optimum overlap
Clustering / subset	754	707/754	0/754
Routing / edge	36	19/36	0/36
Routing / position	7	0/7	0/7

Every target and native minimum passed CPLEX and exhaustive checks.

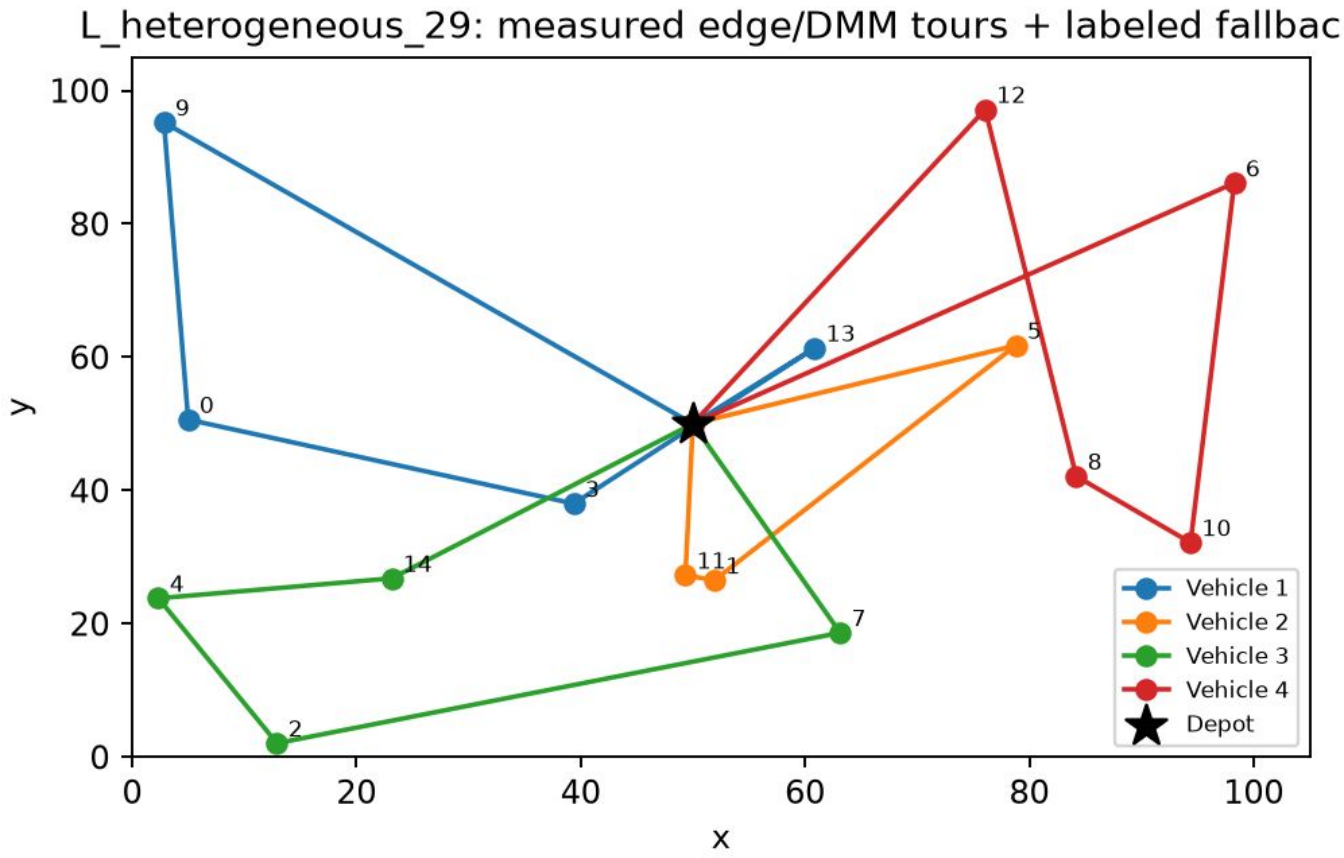
Edge banks return valid sampled tours in 36/36 cases.
Position banks return valid sampled tours in 3/7 cases.

Historical Track B encoding comparison



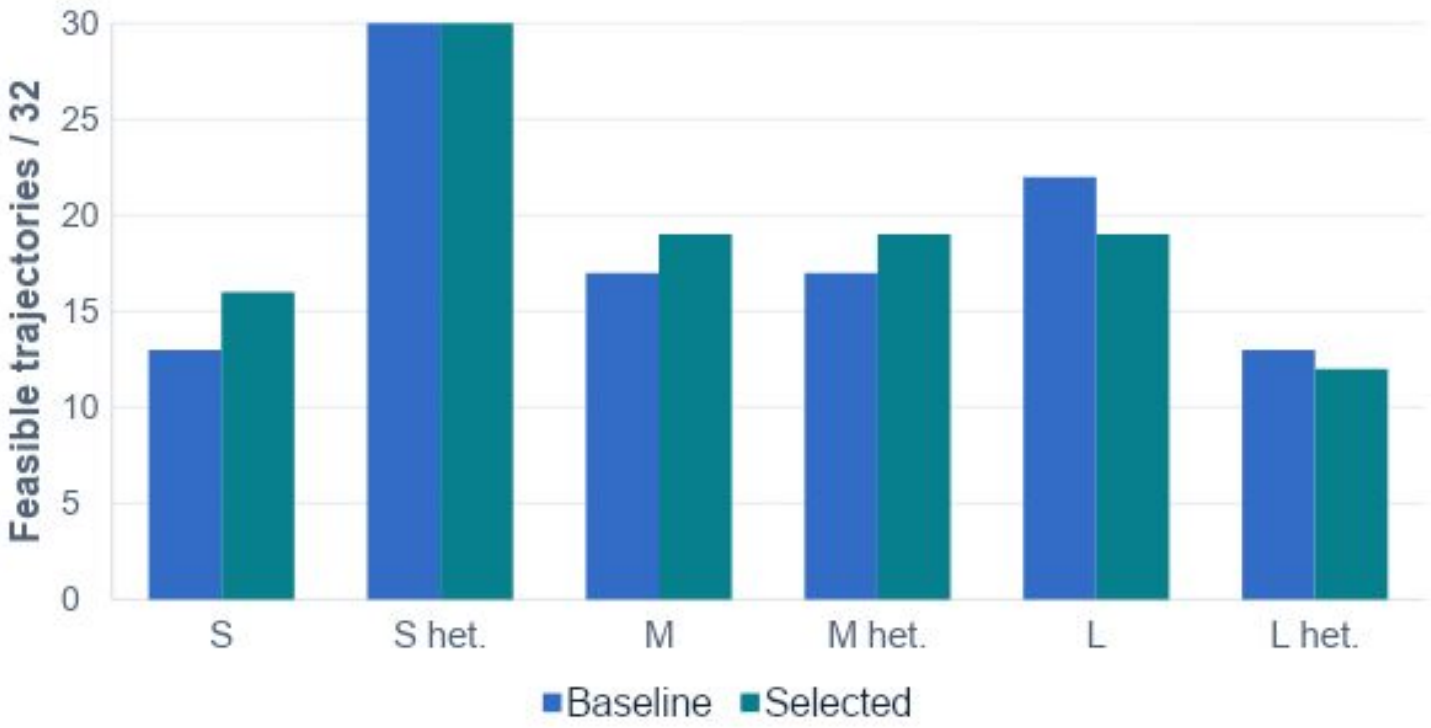
Original GitHub benchmark evidence. Different experiment from the fresh local CPLEX runs.

Historical Track B route example



Quantum routing applies to eligible four-customer groups. Trivial and size fallback groups remain classical.

Sampler tuning on six held-out cases



58.85% to 60.42%

Mean feasible trajectory fraction

8.41% to 7.57%

Mean best clustering-model gap

18.59 s to 43.42 s

Total held-out solver time

16 settings. Seed 11 training, seed 101 validation. Track A only. Separate from the latest Orbit run.

Evaluation metrics and loss accounting

Full-route optimality gap

$$\mathrm{gap}(\%) = 100\left(\frac{C}{C^*} - 1\right)$$

C* is the exact CVRP optimum. Failed runs have an undefined gap.

Where excess travel enters

$$C_{\mathrm{end}} - C_{\mathrm{ref}} = \underbrace{C_{\mathrm{model}} - C_{\mathrm{ref}}}_{\text{decomposition}} + \underbrace{C_{\mathrm{shot}} - C_{\mathrm{model}}}_{\text{clustering}} + \underbrace{C_{\mathrm{end}} - C_{\mathrm{shot}}}_{\text{routing}}$$

Absolute distances add. The clustering term can be negative because the objectives differ.